

Weird Algorithm

Consider an algorithm that takes as input a positive integer n . If n is even, the algorithm divides it by two, and if n is odd, the algorithm multiplies it by three and adds one. The algorithm repeats this, until n is one. For example, the sequence for $n = 3$ is as follows:

3 10 5 16 8 4 2 1

Your task is to simulate the execution of the algorithm for a given value of n .

Input

The only input line contains an integer n .

Input: 3

Output: 3 10 5 16 8 4 2 1

Approach

The problem can be solved by running a while loop till N is not equal to 1. Inside the while loop, check if N is odd then multiply it by 3 and add 1 to it otherwise if N is even then divide it by 2.